Unit - III

V = Set of vertices

E = Set of edges

Graph Theory

$G = (V, E)$ $\quad V \neq \phi$

E may be ϕ.

Graph:

A graph is defined by an ordered pair $G = (V, E)$ or (V, E) where V = Set of all vertices which is a non-empty set and E = set of all edges (possibly empty) such that each edge is assigned an ordered pair of vertices.

Vertices are also called points, nodes or dots.

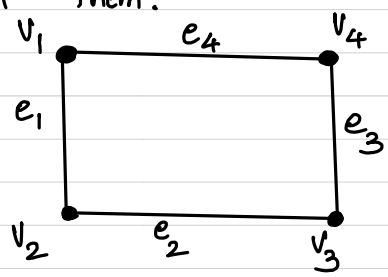
If e is an edge with end vertices u and v, then e is said to be connect or Join u and v

Notation: $e \leftrightarrow (u, v)$

Adjacent Vertices:

Two vertices u and v are said to be adjacent to each other if there is an edge between them.

E.g

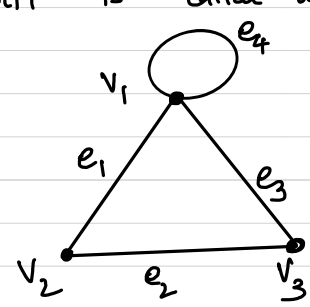


Here v_1 and v_2 are adjacent but v_1 and v_3 are not adjacent.

Loop:

In a graph G, an edge joining a vertex to itself is called a loop.

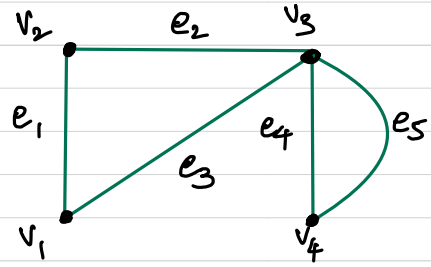
E.g



$\rightarrow$ e_4 is a loop

Parallel edges :

In a graph G, if two vertices are joined by more than one edges then they are called parallel edges / multiple edges.

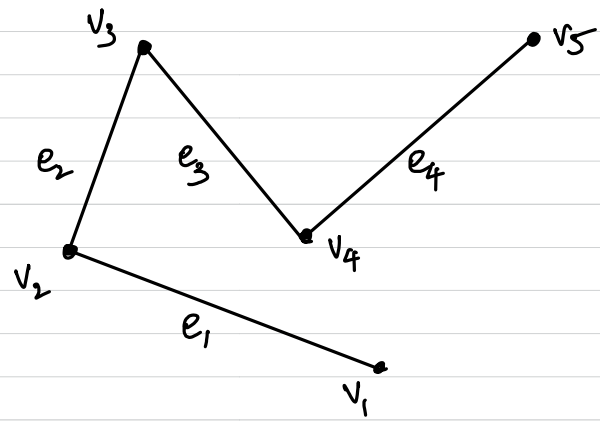


$\longrightarrow$ e_4 and e_5 are parallel edges.

Simple graph :

A simple graph G is a graph without loop and without parallel edges.

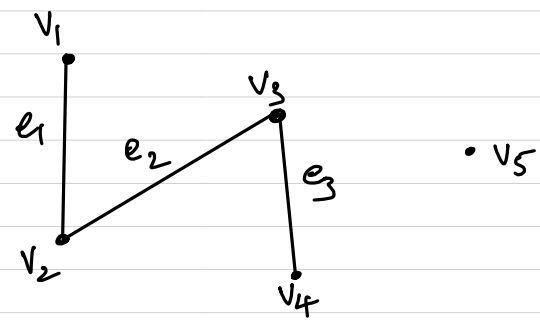
E.g



Isolated vertex :

A vertex is said to be an isolated vertex if it is not the end vertex of any edge of a graph.

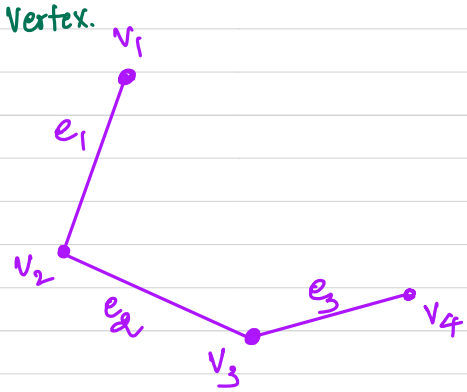
E.g



Here, v_5 is an isolated vertex.

Pendant Vertex :

If a vertex in a graph has only one edge incident to it, then it is called a pendant vertex.

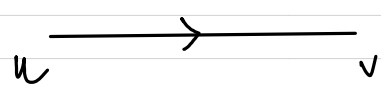


V_1 and V_4 are pendant vertices.

Digraph :

In a digraph / directed graph for the ordered pair of vertices (u, v), the line segment between u and v i.e. the arrow between u and v is represented with an arrow directed from u to v.

E.g



u is called a starting vertex / initial vertex.

v is called a ending vertex / terminal vertex.

Out degree :

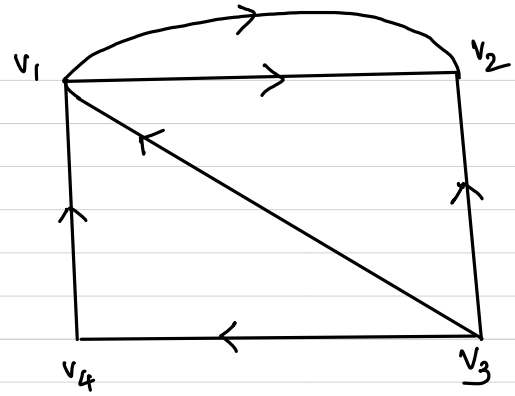
For digraphs, the number of edges having a vertex v as their initial vertex is called the outdegree of the vertex v.

Indegree :

For digraphs, the number of edges having a vertex v as their terminal vertex is called the indegree of the vertex v.

Notation : indegree : $d^-(v)$ / $id(v)$

outdegree : $d^+(v)$ / $od(v)$

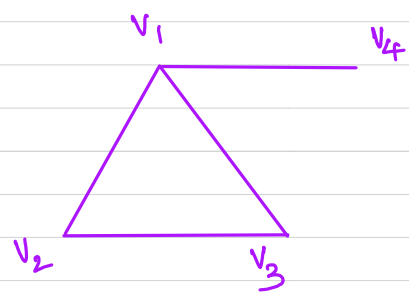


$od(v_1) = 2$	$id(v_1) = 2$
$od(v_2) = 0$	$id(v_2) = 3$
$od(v_3) = 3$	$id(v_3) = 0$
$od(v_4) = 1$	$id(v_4) = 1$

Degree of a vertex in a graph:

In a graph G, the degree of a vertex is the total number of edges incident to it.

E.g



$deg(v_1) = 3$
$deg(v_2) = 2$
$deg(v_3) = 2$
$deg(v_4) = 1$

Hand-Shaking lemma / First theorem in a graph Theory

Statement: In a graph G, the sum of degrees of all vertices is an even number which is equal to twice the number of edges.

or

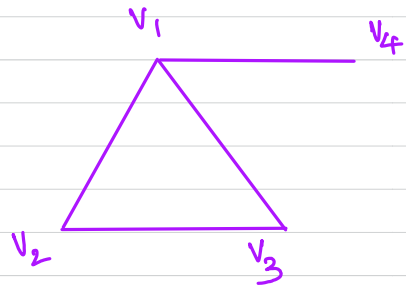
Suppose G is a graph with n vertices $v_1, v_2, \ldots, v_n$. Then

$$\sum_{i=1}^{n} deg(v_i) = 2e.$$

proof When we count the vertex degree, we consider the no. of edges incident on it. Hence every edge have been counted twice.

$$\therefore \sum_{i=1}^{n} \deg(v_i) = 2e$$

E.g



$\deg(v_1) = 3$
$\deg(v_2) = 2$
$\deg(v_3) = 2$
$\deg(v_4) = 1$

$\therefore$ Sum of degrees : $\deg(v_1) + \deg(v_2) + \deg(v_3) + \deg(v_4)$

$= 3 + 2 + 2 + 1$

$= 8$

$= 2(4)$

$\downarrow$

No. of edges.

Even | odd degree vertex : A vertex is said to be even or odd depending upon whether the total degree is an even or odd.

Weighted graph: A graph in which weights are assigned to every edge is called a weighted graph.

Ex Prove that in any graph G, there is an even number of odd vertices.

Soln: Let W = the set of all even vertices

$$= \{ w \mid \deg(w) = \text{even} \}$$

V = the set of all odd vertices

$$= \{ v \mid \deg(v) = \text{odd} \}$$

e = No. of edges.

By Hand-shaking lemma,

$$\sum_{i=1}^{n} \deg(v_i) = 2e$$

$$\therefore \sum_{w \in W} \deg(w) + \sum_{v \in V} \deg(v) = 2e$$

$$\therefore \sum_{v \in V} \deg(v) = \underbrace{2e - \sum_{w \in W} \deg(w)}_{\text{even}}$$

$\boxed{3+7+5}$

$3+7+5+3$

$$\therefore \sum_{v \in V} \deg(v) = \text{even}$$

$\therefore$ We must have even number of terms in $\sum_{v \in V} \deg(v)$

$\therefore$ Even number of vertices in W.

$\therefore$ We have even number of odd vertices.

Isomorphic graphs :

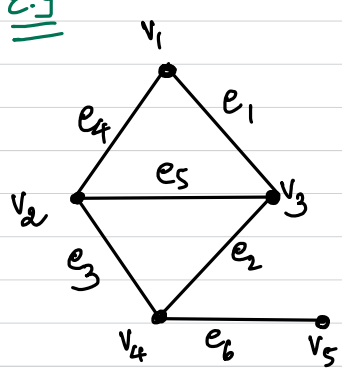
Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is said to be isomorphic if

(i) The number of vertices in V_1 and V_2 are same.

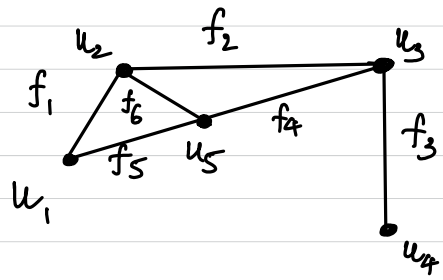
(ii) The number of edges in E_1 and E_2 are same.

(iii) The incidence relationship between the vertices and edges in G_1 and G_2 is completely maintained.

E.g



G_1



G_2

In G_1 and G_2, there are five vertices

In G_1 and G_2, there are six edges.

$v_1 \leftrightarrow u_1$

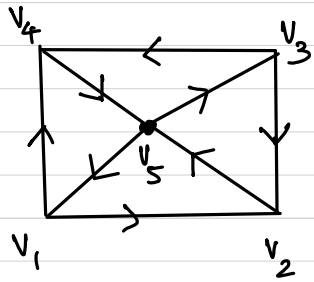
$v_2 \leftrightarrow u_2$

$v_3 \leftrightarrow u_5$

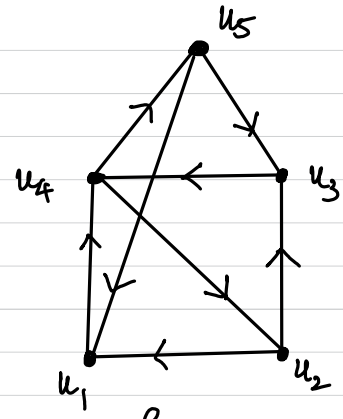
$v_4 \leftrightarrow u_3$

$v_5 \leftrightarrow u_4$

E.g



G_1



G_2

$v_1 \to u_5$

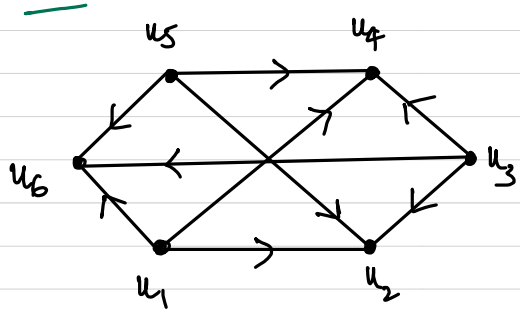
$v_2 \to u_1$

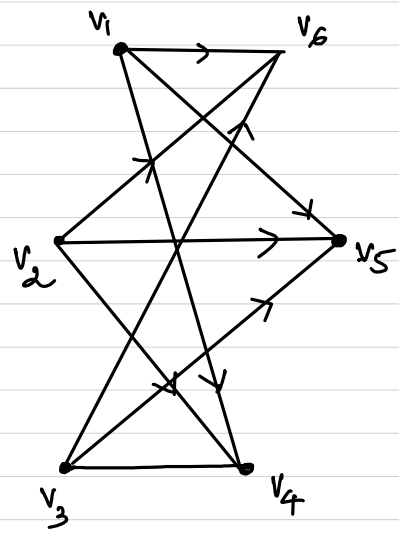
$v_3 \to u_2$

$v_4 \to u_3$

$v_5 \to u_4$

E.g





Subgraph :

Let H be a graph with vertex set $V(H)$ and edge set $E(H)$ and similarly, let G be a graph with vertex set $V(G)$ and edge set $E(G)$. If $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$, then we say that H is a subgraph of G.

If H is a subgraph of G, we write $H \subseteq G$.

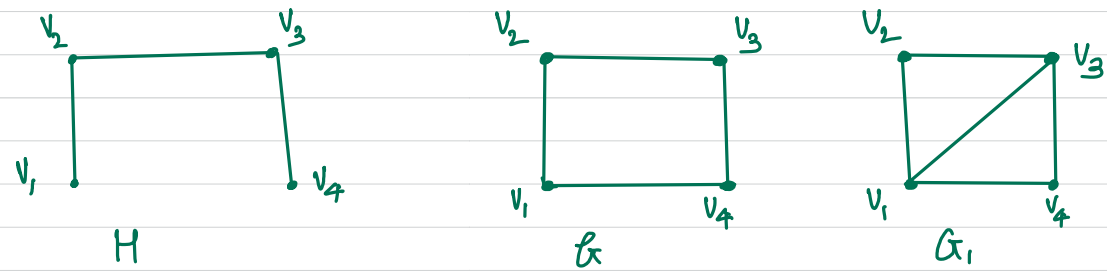
When $H \subseteq G$ but $H \neq G$ i.e. $V(H) \neq V(G)$ or

$E(H) \neq E(G)$ then H is called a proper subgraph of G.

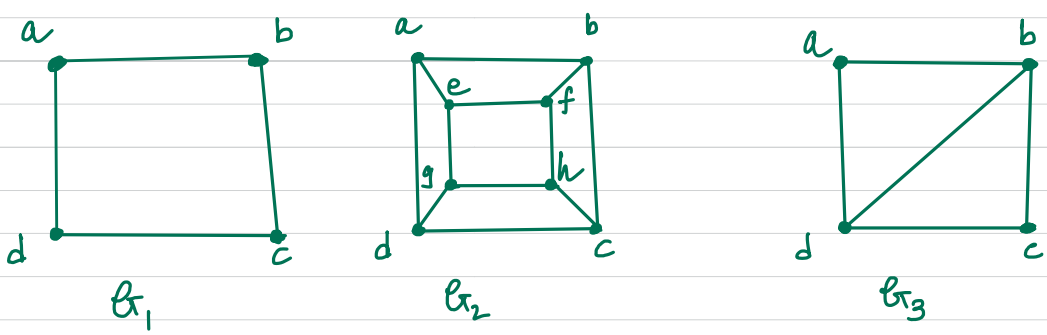
Spanning Subgraph :

A spanning subgraph of G is a subgraph of H with $V(H) = V(G)$

i.e., H and G have exactly the same vertex set.



Here, $H \subset G \subset G_1$.



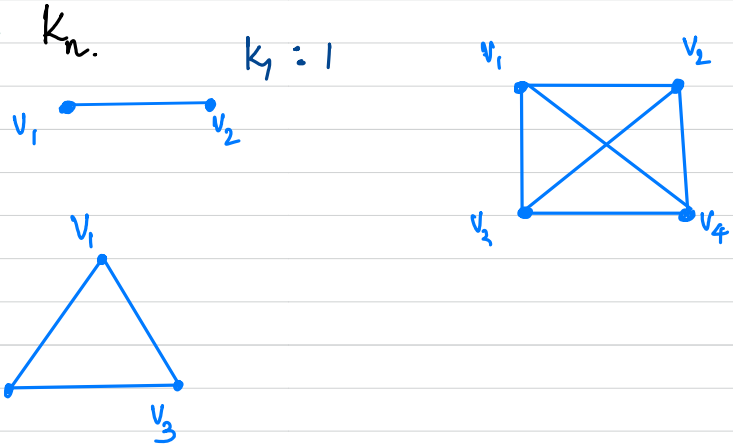
Here, $G_1 \subseteq G_2$, $G_1 \subseteq G_3$ but $G_3 \nsubseteq G_2$.

Complete Graph :

A simple graph G is said to be complete if there is an edge between each pair of vertices.

Notation : K_n.

E.g



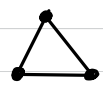
Note :

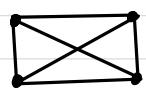
→ In a complete graph K_n, there are $\frac{n(n-1)}{2}$ edges.

K - regular graph :

If for some positive integer k, the degree of each vertex of a graph is k, then it is said to be k - regular.

1 - regular : K_1

2 - regular :  K_2

3 - regular :  K_3.

In general, K_n is $(n-1)$ regular

Ex Draw two 3- regular graph with eight vertices.

Sol^n : Try

Ex Prove that it is impossible to have a party of 9 people such that each one knows exactly five of others.

Sol^n : The persons in a party represents vertices say $v_1, v_2, v_3, \ldots, v_9$. If two persons know each others then we draw an edge between them.

Suppose each one knows exactly five of others

$\therefore$ degree of every vertex will be 5.

$\therefore$ The sum of all the degrees of vertices : 5×9

But by Hand- shaking lemma,

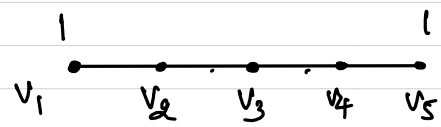
$$5 \times 9 = 2e$$

$$\therefore e = \frac{45}{2}$$

Which is a contradiction to the fact that the number of edges is a natural number.

$\therefore$ It is impossible to have a party of nine people where each person knows exactly five.

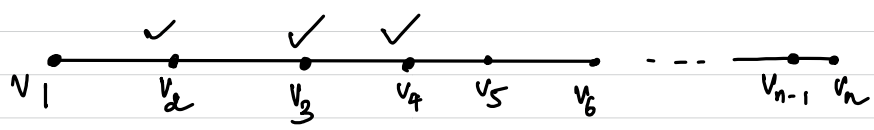
Ex : Find the number of edges of a linear graph (straight line graph) on n vertices.



$$1+1+2+2+2 = 2e$$

$$8 = 2e$$

$$e = 4$$



Sum of degrees $= 2e$

$\therefore$ $\deg(v_1) + \deg(v_n) +$ deg of $n-2$ vertices $= 2e$

$\therefore$ $1 + 1 + 2(n-2) = 2e$

$\therefore$ $2 + 2n - 4 = 2e$

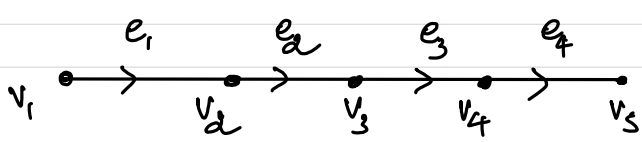
$\therefore$ $2n - 2 = 2e$

$e = n-1$

Path and Circuits :

Path :

In a directed graph, a path is a sequence of edges $(e_1, e_2, \ldots, e_k)$ such that the terminal vertex of the sequence e_i is the initial vertex of some e_j, appearing next in the sequence where $1 \le i$, $j \le k$.



Length of a path :

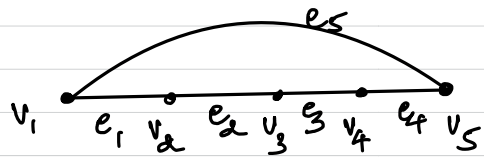
The number of edges appearing in the sequence of a path is called the length of the path.

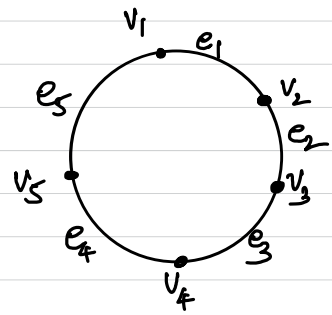
Elementary Path :

A path is said to be an elementary path if it does not meet the same vertex twice.

Circuit | Cycle :

A path which originates and ends in the same vertex is called a Circuit or Cycle.





$v_i - v_j$

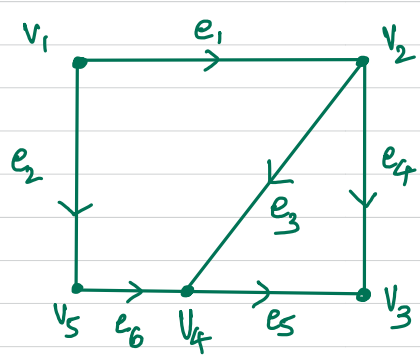
Acyclic :

A simple digraph which does not have any cycle is called acyclic.

distance between vertices : Length of shortest path.

Examples :

(1)



For the above graph

(a) Find all the elementary path from v_1 to v_3.

(b) Find $d(v_1, v_3)$.

Sol^n :

(a) Elementary path : No vertex is repeated.

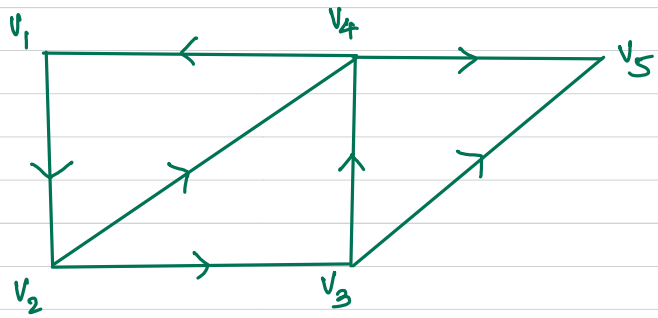
$\therefore$ Paths are : $v_1 - v_2 - v_3$

$v_1 - v_2 - v_4 - v_3$

$v_1 - v_5 - v_4 - v_3$.

(b) $d(v_1, v_3) = 2$.

(2) Find all indegree and outdegree of the vertices of the graph given below. Give all elementary cycles.



Vertex	Indegree	Outdegree	Total degree
v_1	1	1	2
v_2	1	2	3
v_3	1	2	3
v_4	2	2	4
v_5	2	0	2

Elementary cycles : $v_1 v_2 v_4 v_1$, $v_1 v_2 v_3 v_4 v_1$

Find list of all the vertices which are reachable from another vertex of the digraph.

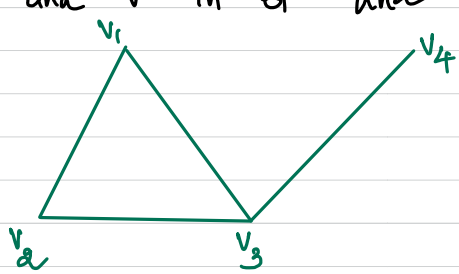
Reachable vertices :

A vertex v_j is said to be reachable from a vertex v_i if there is a path between them.

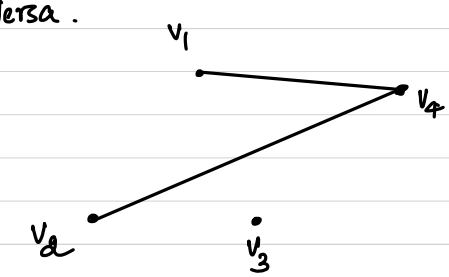
Vertex	Reachable set of vertices
v_1	v_2, v_3, v_4, v_5, v_1
v_2	v_1, v_3, v_4, v_5, v_2
v_3	v_1, v_2, v_4, v_5, v_3
v_4	v_1, v_2, v_3, v_5, v_4
v_5	v_5

Complement of a graph :

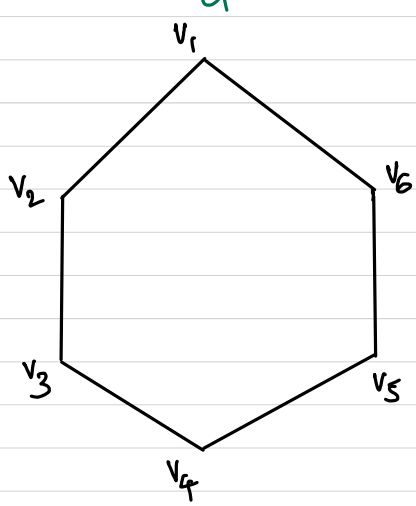
Let G be a simple graph. The complement of a graph G denoted by $\overline{G}$ is a simple graph in which if there is an edge between u and v in G, then there is no edge between u and v in $\overline{G}$ and vice versa.



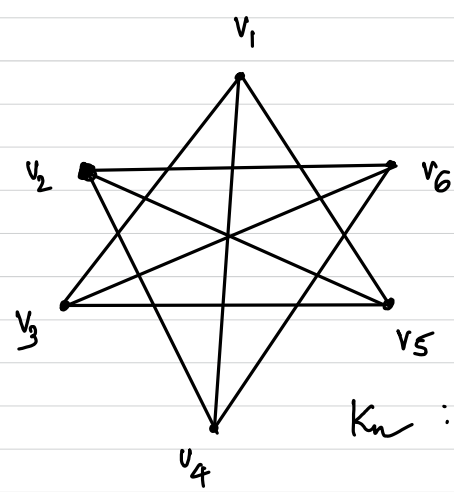
G



$\overline{G}$



G



$\overline{G}$

K_n :

$\frac{n(n-1)}{2}$

Note :

If we combine all the edges of G and $\overline{G}$ we get the number of edges of a complete graph

i.e No. of edges of G + No. of edges of $\overline{G}$ $= \dfrac{n(n-1)}{2}$

No. of edges in K_n

Complement of a digraph:

The complement of a simple digraph $G = \langle V, E \rangle$ is denoted by $\overline{G}$ and is defined as $\overline{G} = \langle V, \overline{E} \rangle$ where $\overline{E} = V \times V - E$

i.e. (i) The vertex set is same as the edge set

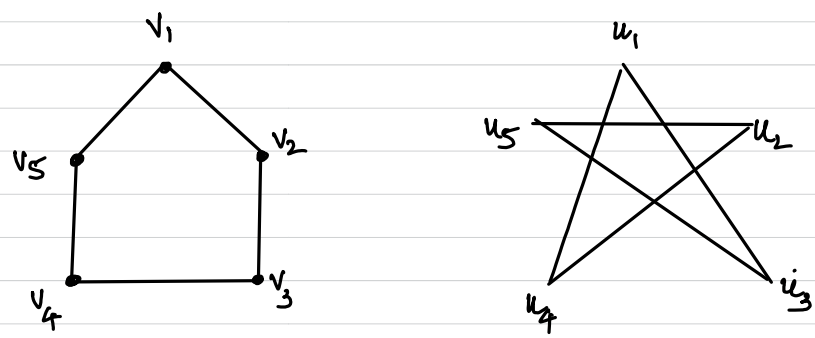
(ii) If u and v are adjacent in G, they are not adjacent in $\overline{G}$.

Self - complementary graph:

A simple graph G is said to be self - complementary if it is isomorphic to its own complement.

i.e A simple graph G is said to be self-complementary if $G \cong \overline{G}$.

E.g



G $\qquad$ $\overline{G}$

Note that, $G = \overline{G}$.

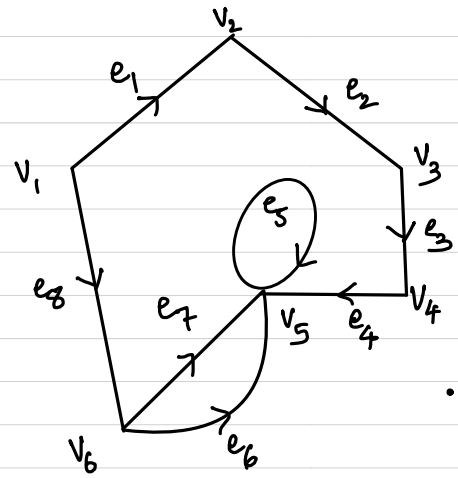
$v_1 \leftrightarrow u_1$	$v_3 \leftrightarrow u_5$	$v_5 \leftrightarrow u_4$
$v_2 \leftrightarrow u_3$	$v_4 \leftrightarrow u_2$	

So, G is a self - complementary graph.

Geodesic :

If a vertex v is reachable from the vertex u, then of all such possible paths, the path of minimum length is called a geodesic.

The length of geodesic is called the distance from u to v.



$P_1 : v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5$

$P_2 : v_1 e_8 v_6 e_7 v_5$

$P_3 : v_1 e_8 v_6 e_6 v_5$

Here, $d(v_1, v_5)$ = minimum of the lengths of all possible paths from v_1 to v_5.

$= 2.$

Note :

If a vertex v_j is not reachable from a vertex v_i, we write $d(v_i, v_j) = \infty$.

E.g. $d(v_5, v_6) = \infty$

Eccentricity :

The eccentricity of a vertex u, defined as $E(u)$, is the maximum of its distances to other vertices.

i.e., $E(u) = \max\limits_{v_j \in G} d(u, v_j)$

Diameter :

The diameter or diam G is the maximum of the vertex eccentricities.

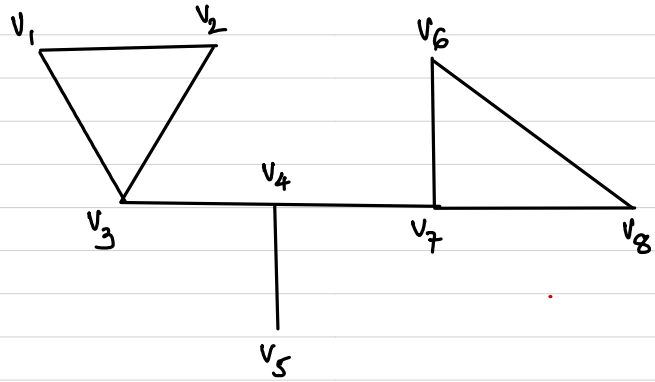
$$\text{diam } G = \max_{u,v \in G} d(u,v)$$

Radius :

Radius or rad G is the minimum of the Vertex eccentricity.

$$\text{rad } G = \min_{u,v \in G} d(u,v).$$

E.g



We first find the distances of all vertices from v_1

$d(v_1, v_2) = 1$	$d(v_1, v_6) = 4$
$d(v_1, v_3) = 1$	$d(v_1, v_7) = 3$
$d(v_1, v_4) = 2$	$d(v_1, v_8) = 4$
$d(v_1, v_5) = 3$	$\epsilon(v_1) = 4$

$$\max d(v_1, v_i) = \text{eccentricity of } v_1$$
$$= \epsilon(v_1)$$
$$= 4$$

Similarly, we find eccentricities of v_2 to v_8.

$\epsilon(v_2) = 4$	$\epsilon(v_4) = 2$	$\epsilon(v_6) = 4$	$\epsilon(v_8) = 4$.
$\epsilon(v_3) = 3$	$\epsilon(v_5) = 3$	$\epsilon(v_7) = 3$	

Now,

Rad G = min. of eccentricities of all the vertices = 2

Diam G = max of eccentricities of all the vertices = 4.

Connected Vertices :

Let G be a graph. A vertex u in a graph G is said to be connected with a vertex v if there exists a path between u and v.

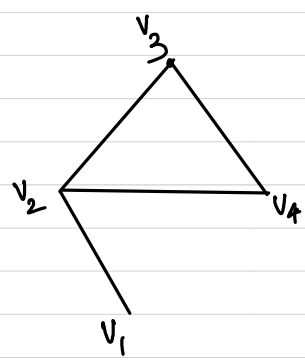
Connected graph :

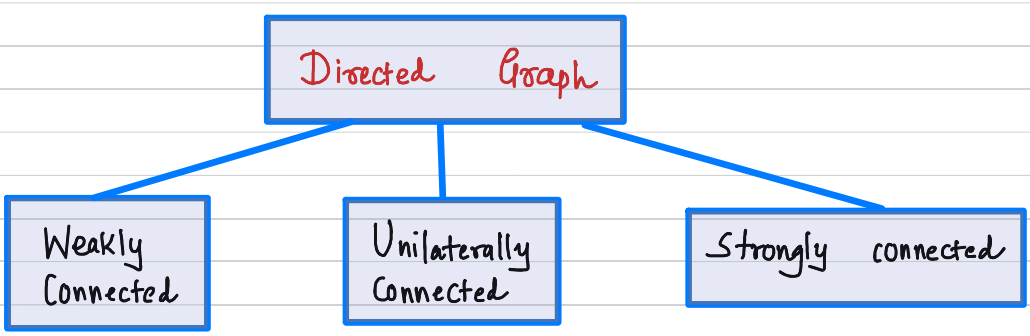
An undirected graph G is connected if for any two vertices u and v, there is a path from u to v.

<u>or</u>

An undirected graph G is connected if any two vertices are connected.

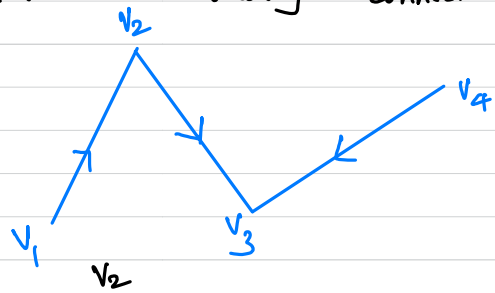
<u>E.g</u>

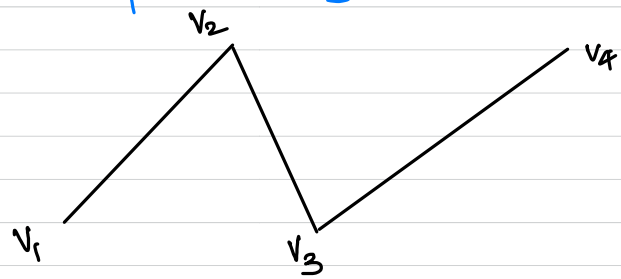




Weakly connected Graph :

If the undirected graph obtained by neglecting the directions of the edges of the given graph is connected, then the digraph is called weakly connected graph.





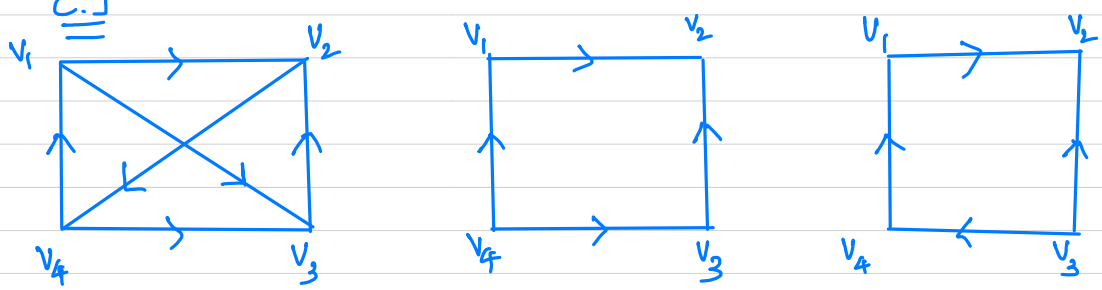
Unilaterally Connected :

A digraph G is said to be unilaterally connected if given any pair of vertices u and v of the digraph, either u is reachable from v or v is reachable from u but not both.

Strongly Connected graph :

A digraph D is said to be strongly connected if given any pair of vertices u and v of the digraph, each one is reachable from the other.

E.g



(a) (b) (c)

$\boxed{\text{Strongly connected} \Rightarrow \text{Unilaterally Connected} \Rightarrow \text{Weakly connected}}$

Sol^n : (a) strongly connected

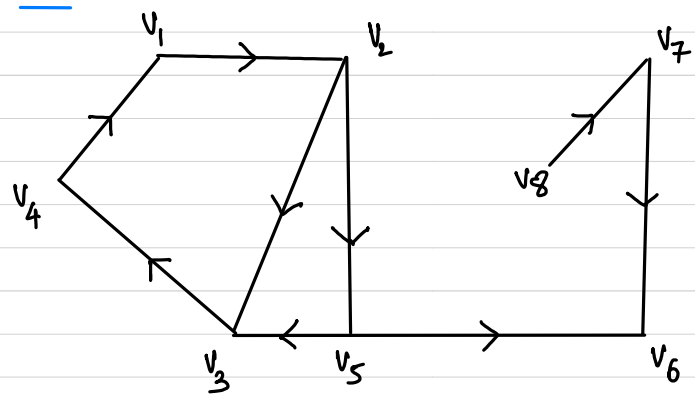
(b) Only weakly connected

(c) Unilaterally connected but not strongly connected as v_1 and v_2 are not reachable from each other.

Subgraph induced by the vertex set :

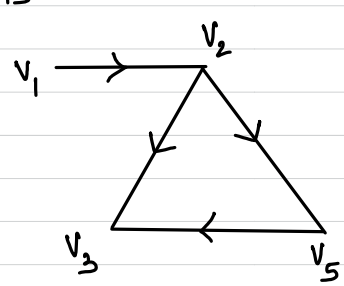
Let $G = (V, E)$ be a simple digraph and $X \subseteq V$. A subgraph whose vertices are given by the set X and whose edges consists of all those edges in G which have their initial and terminal vertices in X is called the subgraph generated by X.

E.g

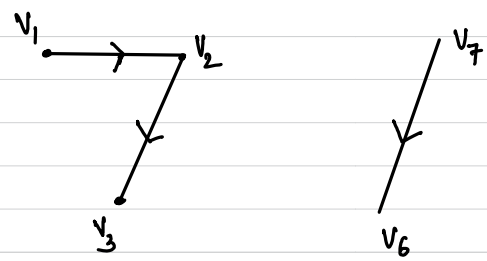


G

Take $X = \{v_1, v_2, v_3, v_5\}$ then the subgraph induced by X is



If $Y = \{v_1, v_2, v_3, v_6, v_7\}$ then the subgraph generated by Y is follows :



Ex Let G be an undirected graph. For vertices u and v, define $u \sim v$ if $u = v$ or if there is a path from u to v. Show that $\sim$ is an equivalence relation on $V(G)$

Sol^n: By definition $u \sim u$; $\forall u \in V(G)$

$\therefore \sim$ is reflexive

Suppose $u \sim v$.

$\Rightarrow$ There is a path P from u to v.

$\Rightarrow$ There is a path from v to u

$\Rightarrow$ $v \sim u$

$\Rightarrow$ $\sim$ is symmetric.

Suppose $u \sim v$ and $v \sim w$.

$\Rightarrow$ There is a path P_1 from u to v

and P_2 from v to w.

$\therefore$ There is a path $P_1 P_2$ from u to w.

$\therefore$ $u \sim w$.

$\therefore$ $\sim$ is transitive

Thus, $\sim$ is an equivalence relation.

Matrix Representation of Graphs :

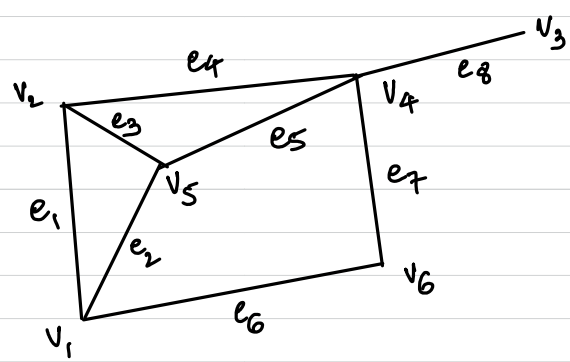
(A) Adjacency Matrix for undirected graph :

The adjacency matrix for an undirected graph G with n vertices is $n \times n$ matrix

$A = [a_{ij}]$ such that

$$a_{ij} = \begin{cases} 1, & \text{if there is an edge between } i^{th} \text{ vertex to } j^{th} \text{ vertex} \\ 0, & \text{otherwise.} \end{cases}$$

E.g

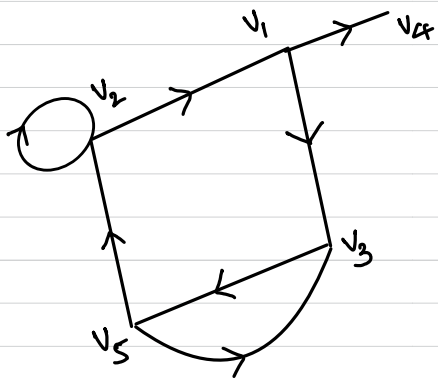


$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}_{6\times 6} \end{array}$$

(B) Adjacency matrix for a directed graph

Let G be a digraph. Then the adjacency matrix $A = (a_{ij})$ of a digraph G is $n \times n$ matrix such that

$$a_{ij} = \begin{cases} 1; & \text{if there is an edge from } i^{th} \text{ vertex to } j^{th} \text{ vertex} \\ 0; & \text{otherwise.} \end{cases}$$

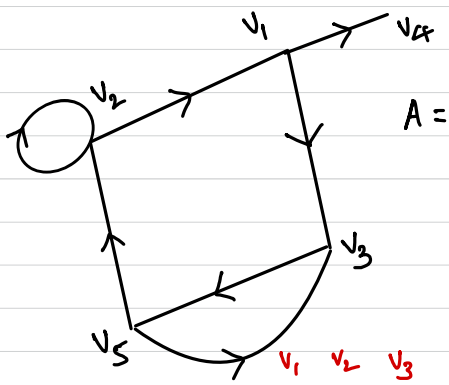


$$\begin{array}{c|ccccc} & v_1 & v_2 & v_3 & v_4 & v_5 \\ \hline v_1 & 0 & 0 & 1 & 1 & 0 \\ v_2 & 1 & 1 & 0 & 0 & 0 \\ v_3 & 0 & 0 & 0 & 0 & 1 \\ v_4 & 0 & 0 & 0 & 0 & 0 \\ v_5 & 0 & 1 & 1 & 0 & 0 \end{array}$$

(c) Path Matrix :

Let G be a simple digraph in which no. of vertices is n and all the vertices are assumed to be ordered. An $n \times n$ matrix P whose elements are given by

$$p_{ij} = \begin{cases} 1 & \text{if there is a path from } v_i \text{ to } v_j \\ 0 & \text{otherwise} \end{cases}$$



$$A = \begin{array}{c|ccccc} & v_1 & v_2 & v_3 & v_4 & v_5 \\ \hline v_1 & 0 & 0 & 1 & 1 & 0 \\ v_2 & 1 & 1 & 0 & 0 & 0 \\ v_3 & 0 & 0 & 0 & 0 & 1 \\ v_4 & 0 & 0 & 0 & 0 & 0 \\ v_5 & 0 & 1 & 1 & 0 & 0 \end{array}$$

$$A^2 = \begin{array}{c|ccccc} & v_1 & v_2 & v_3 & v_4 & v_5 \\ \hline v_1 & 0 & 0 & 0 & 0 & 1 \\ v_2 & 1 & 1 & 1 & 1 & 0 \\ v_3 & 0 & 1 & 1 & 0 & 0 \\ v_4 & 0 & 0 & 0 & 0 & 0 \\ v_5 & 1 & 1 & 0 & 0 & 1 \end{array}$$

$$A^3 = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 1 & 0 \end{bmatrix}$$

$a_{53} = 2$

$v_5 \quad v_3$:

$$A^4 = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 \\ 1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 1 & 1 & 2 \end{bmatrix}$$

$V_5 \quad V_2$

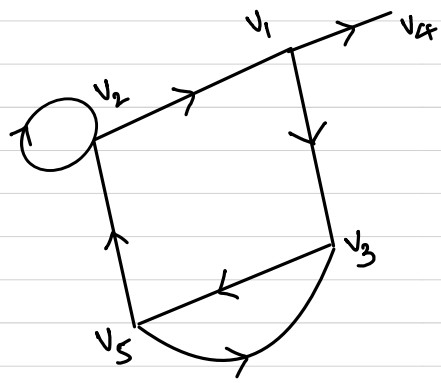
n - vertices :

$n \times n$

$n-1$

$$B^4 = A + A^2 + A^3 + A^4$$

$$B^4 = \begin{bmatrix} 1 & 2 & 2 & 1 & 2 \\ 4 & 5 & 4 & 3 & 2 \\ 2 & 4 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 4 & 6 & 4 & 2 & 3 \end{bmatrix}$$



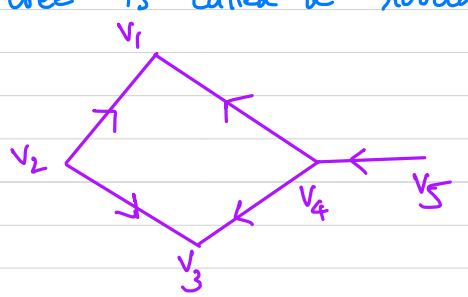
$$P = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Tree :

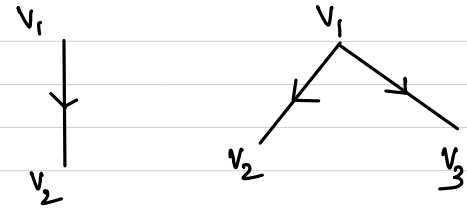
A connected graph without cycle is called a tree.

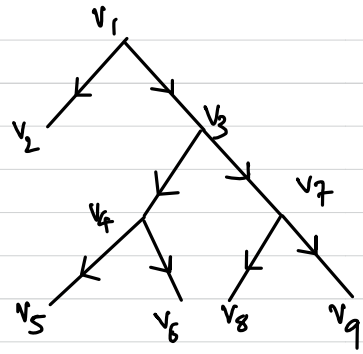
A directed tree is a graph without cycle which has one vertex called root with indegree 0, while all other vertices having indegree 1.

Such a tree is called a rooted tree.

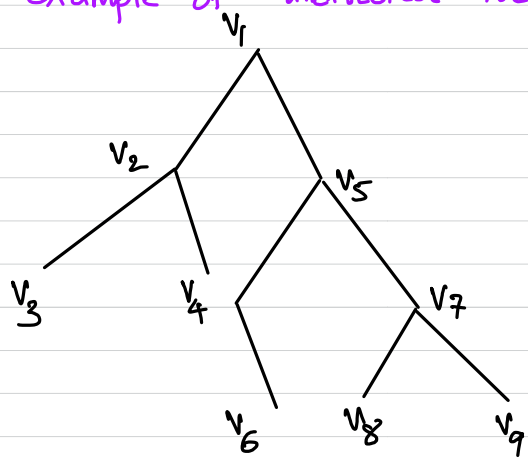


E.g





Example of undirected Tree

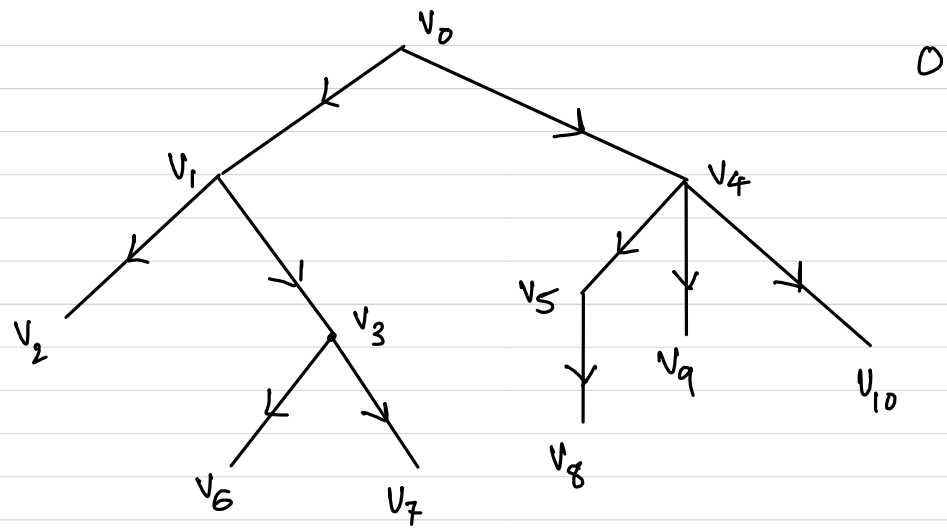


Level of a Vertex :

In a directed tree, the level of any vertex is the length of its path from the root.

Level of any vertex is equal to its distance from the root.

Note:- Level of a root vertex in any tree is always zero.



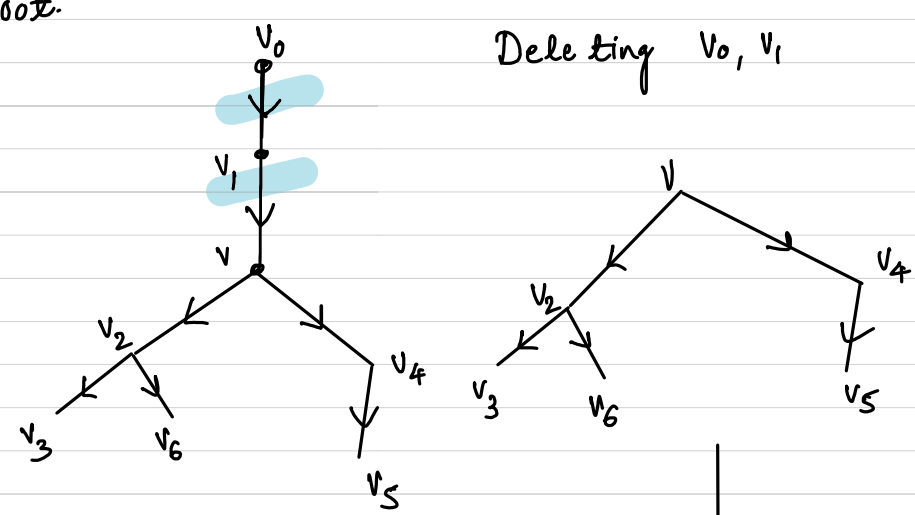
Level of v_0 : 0	Level of v_2 : 2
Level of v_8 : 3	Level of v_{10} : 2
Level of v_6 : 3	Level of v_7 : 3

Ordered tree :

If in a directed tree, an ordering of the nodes at each level is done, such a tree is called an ordering tree.

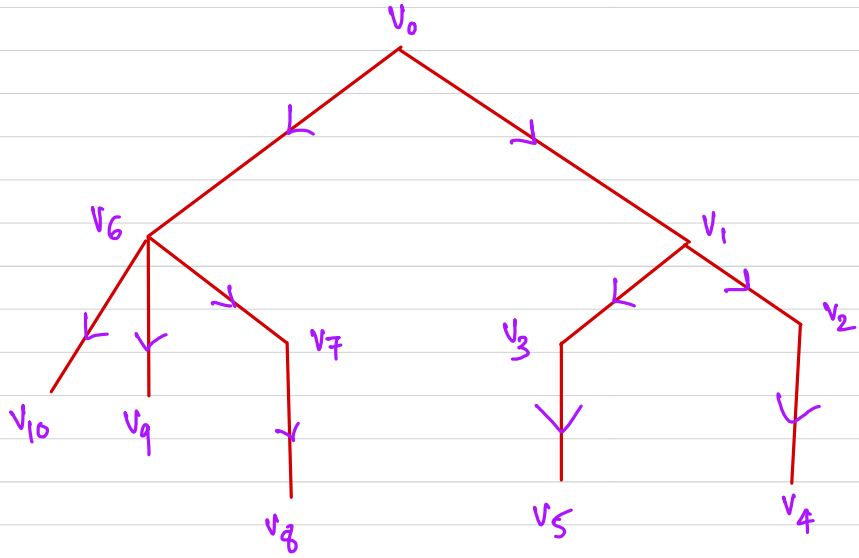
Sub tree :

Let T be a given tree. Let v be a particular vertex of T. By deleting the root of T and the edges joining the root to a vertex v, we get a subtree of T with v as a root.

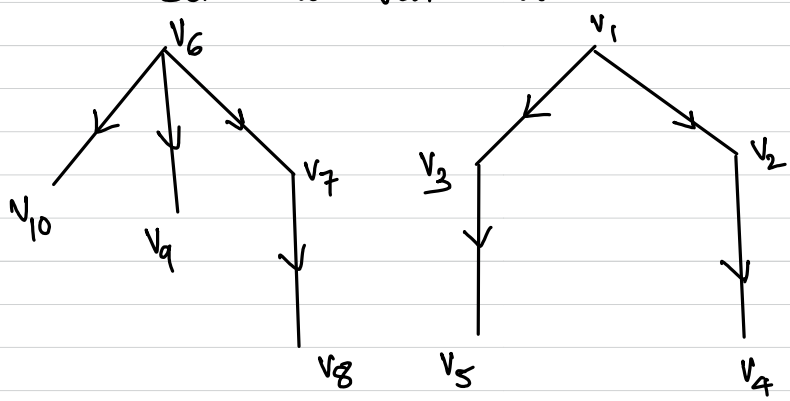


T

Subtree of T.



Delete a vertex v_0.



Forest :

A set of disjoint trees is called a forest.

Def^n :

Let v be a branch vertex in a rooted tree. A vertex u is said to be a son (offspring) of v if there is an edge from v to u.

Also, v is said to be the father (parent) of u.

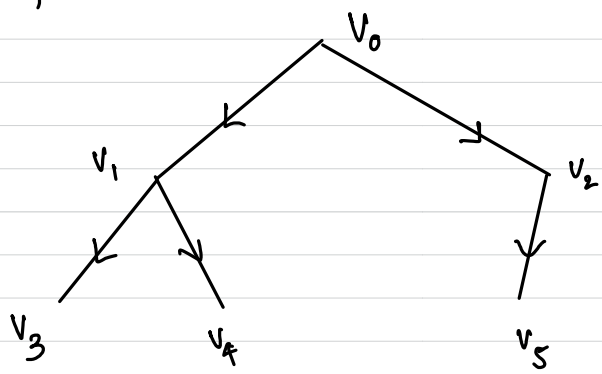
Brothers :

Two vertices are said to be brothers if they are son of the same vertex.

Descendant :

A vertex u is said to be descendant of v if there is a direct path from v to u.

Also, v is called an Ancestor of u.



- V_1 and V_2 are sons of V_0
- V_1 is the father of V_3 and V_4.
- V_3 and V_4 are brothers.
- V_5 is the descendant of V_2.

Recursive definition of directed tree :

A tree contains one or more vertices such that one of the vertex is called the root while all other vertices are partitioned into a finite number of trees called subtrees.

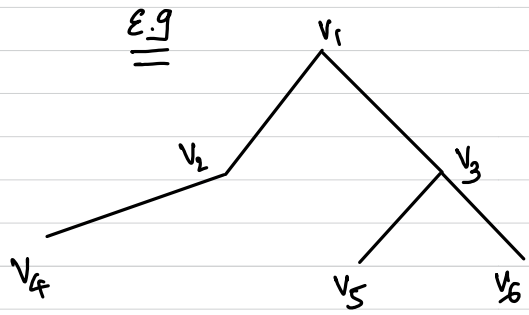
m- ary tree :

If in a directed tree, the outdegree of every vertex is less than or equal to m, then the tree is called m- ary tree.

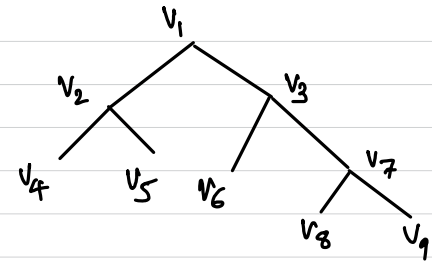
Full | Complete | Regular m- ary tree :

An m-ary tree is called Complete | regular if every one of its branch vertices has exactly m sons.

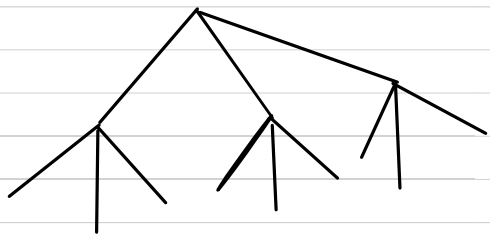
E.g



Binary tree



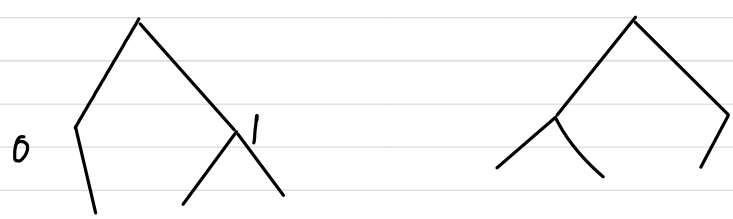
Full Binary Tree

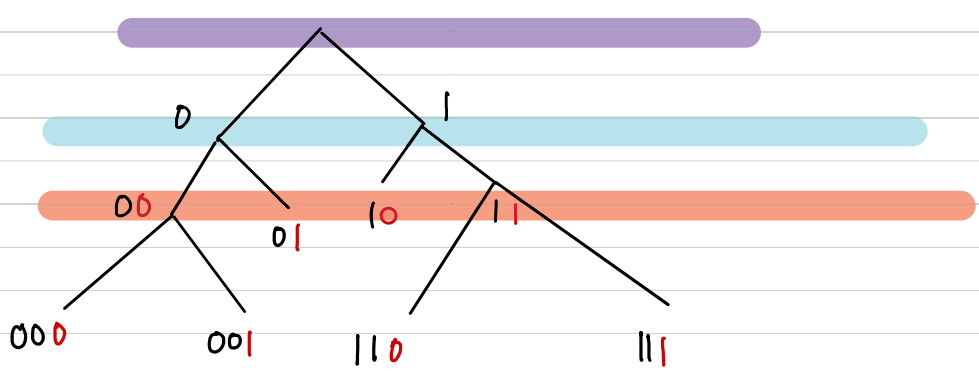


Full Ternary Tree

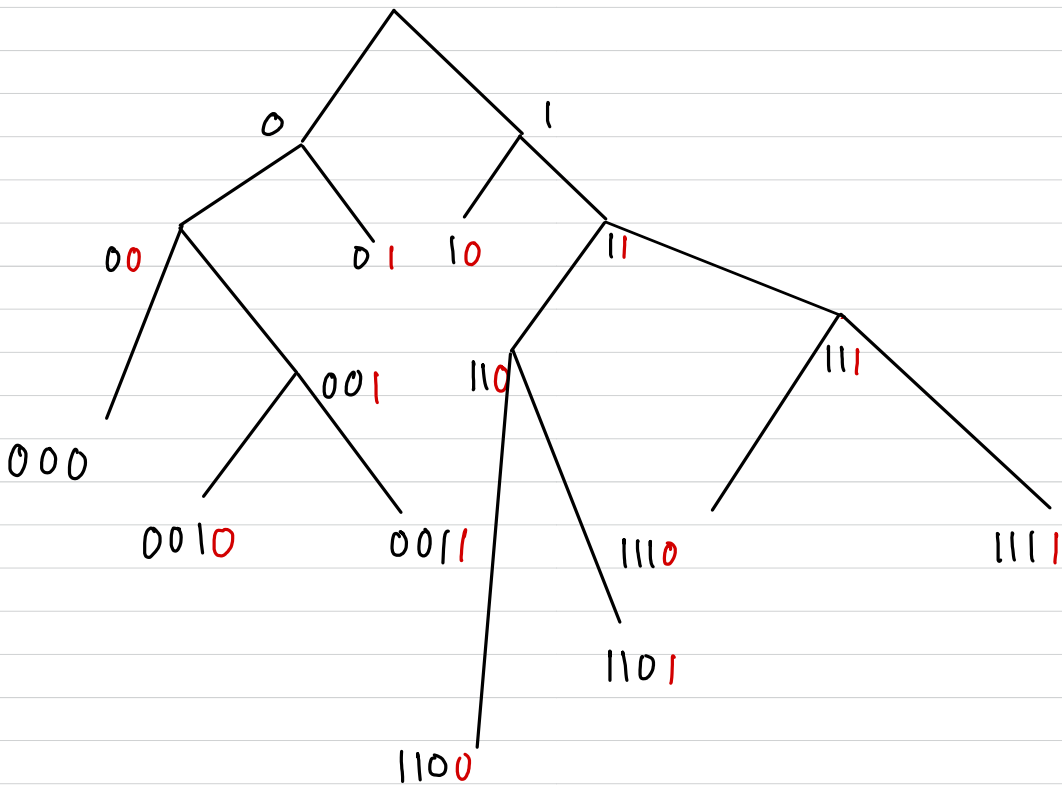
Positional m-ary tree:

An m-ary tree, in which the m (or fewer) sons of any vertex are assumed to have m distinct positions is called a positional m-ary tree



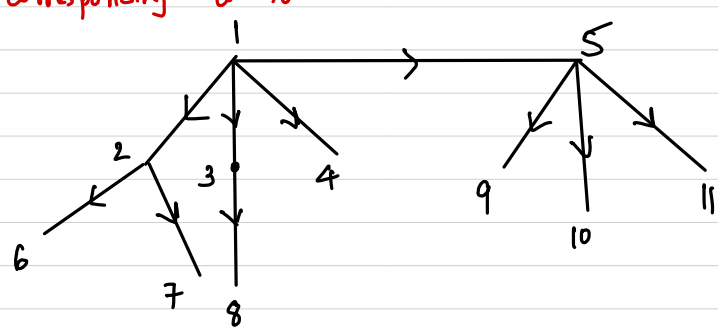


Prefix code : {000, 001, 01, 10, 110, 111}

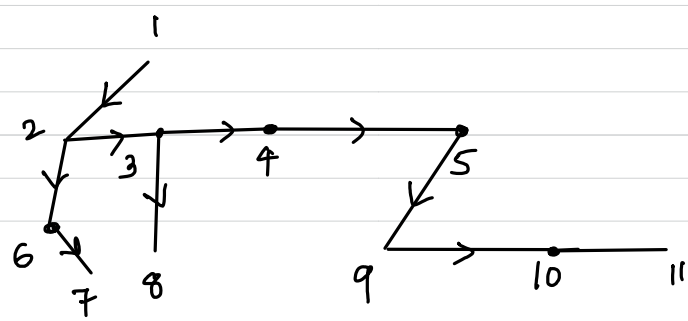


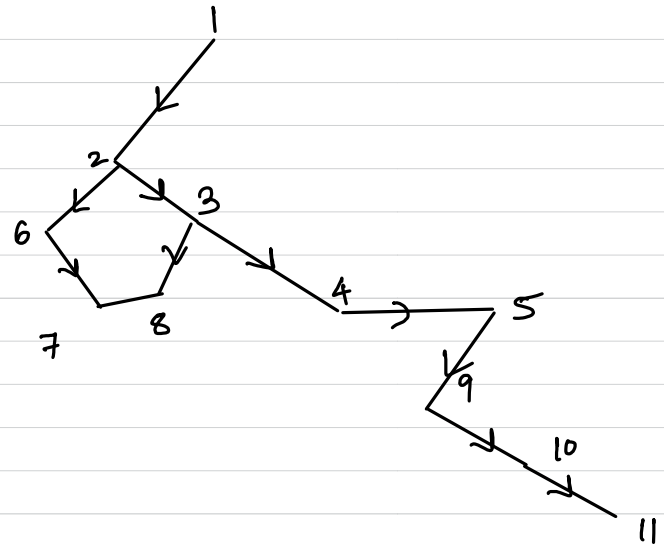
Prefix code : {000, 0010, 0011, 01, 10, 1100, 1101, 1110, 1111}

Ex Given a directed tree, find a binary tree corresponding to it.



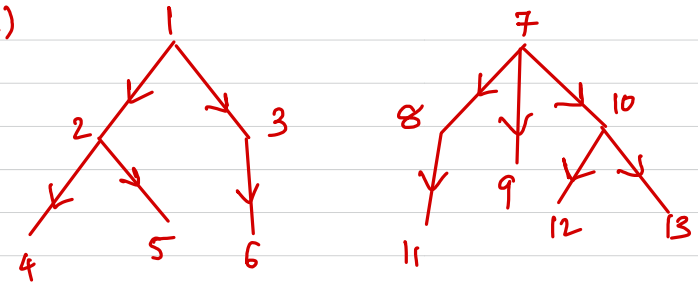
step : 1





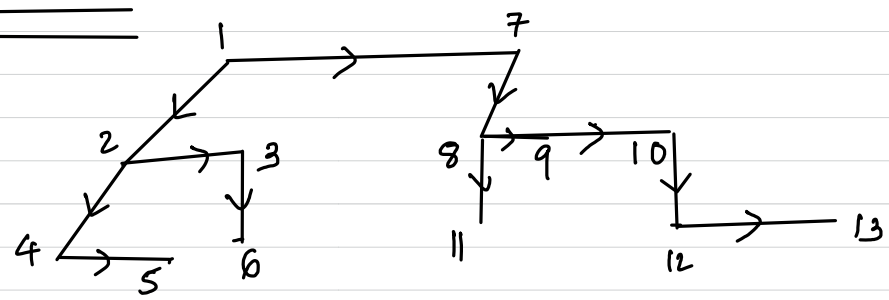
$\boxed{15-20}$

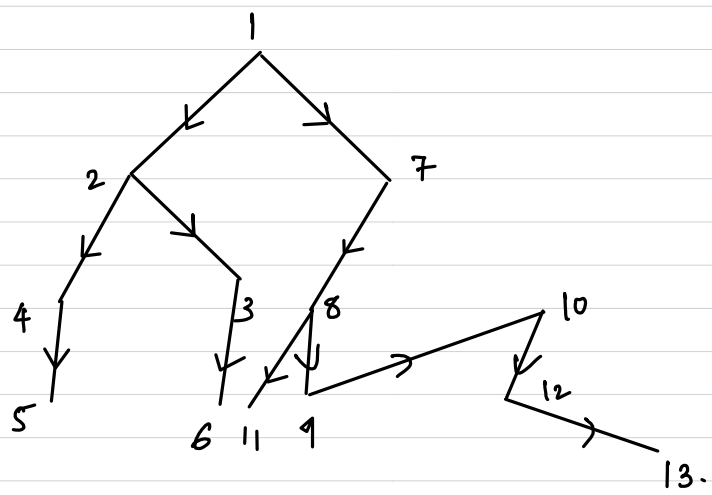
(2)



Minimal

Sol^n Step: I





Spannig Tree:

Let G be a undirected graph. A spanning tree T is a subgraph of G such that number of vertices in G and T are equal.

or

Let G be an undirected graph. A spanning tree T is a subgraph of G with the condition $V(G) = V(T)$.

There are two algorithm to find minimal Spanning Tree

(1) Prims Algorithm

(2) Kruskal Algorithm.

Prims Algorithm

Step I : Start with any arbitrary vertex.

Step II : Look for the edges incidenting on a vertex.

Out of all those, choose an edge with a minimum weight.

Step III : Again look for all edges incidenting on present vertex and previously choosen vertex.

Consider an edge with minimum edge so that no cycle arises in a graph.

Step IV : Repeat the procedure untill every vertex is visited.

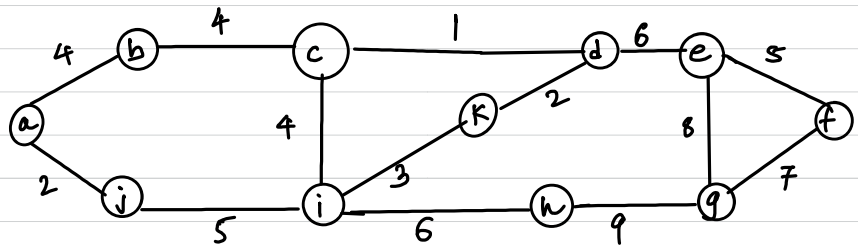
Kruskal Algorithm :

Step I : Choose any edge with minimum weight.

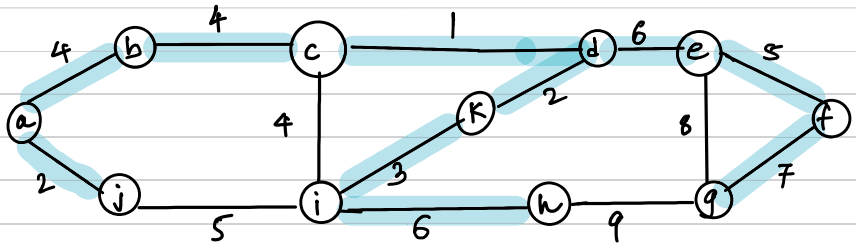
Step II : In the remaining graph choose an edge with the minimum weight so that cycle does not arise.

Step III : Repeat Step II until every vertex of a graph is visited.

(1) Using Prims algorithm, find the minimal spanning tree.



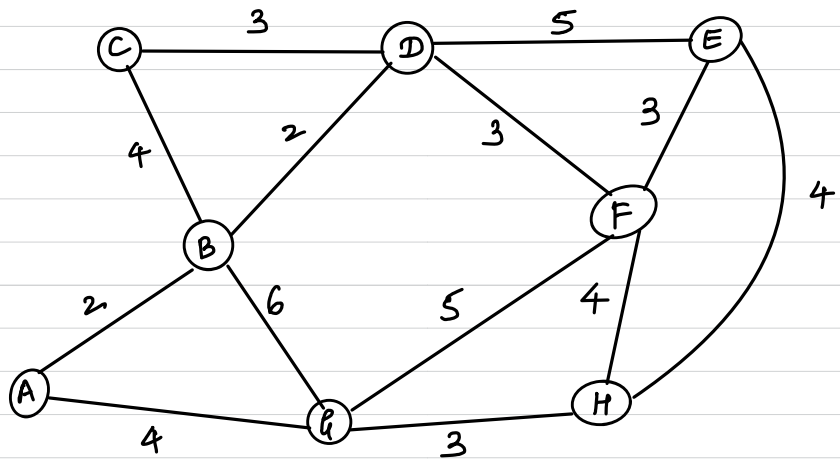
Sol^n :



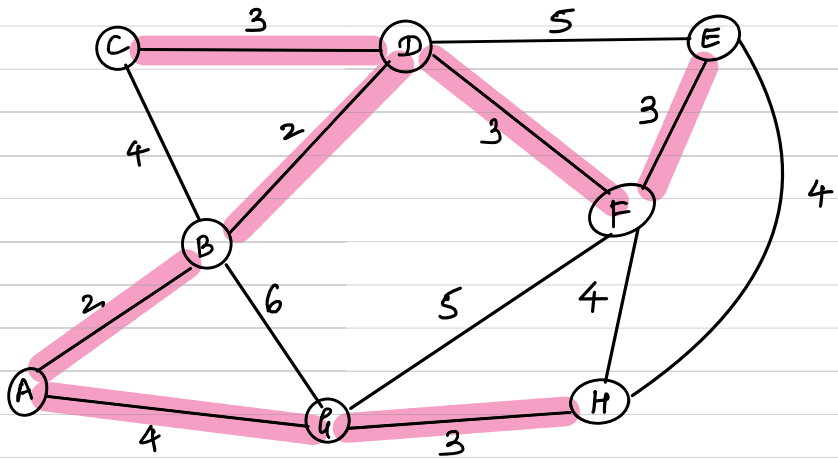
$W(T) = 2+4+4+1+2+3+6+6+5+7$

$= 40$

(2) Find the minimal spanning tree using Prims algorithm.



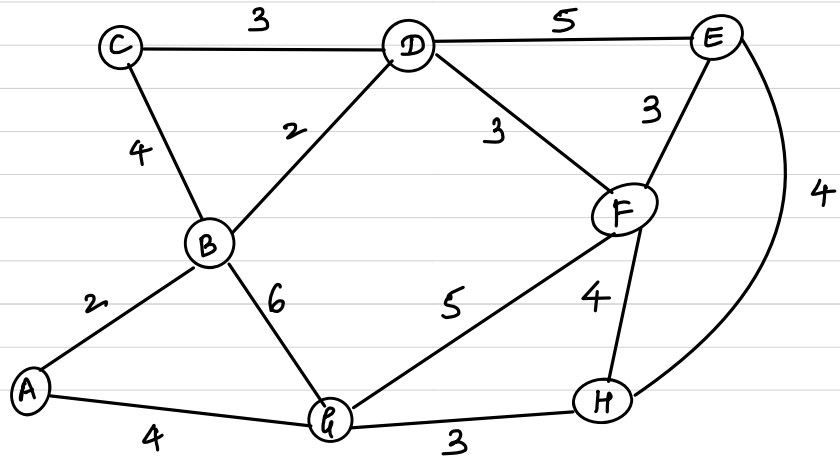
Sol^n :



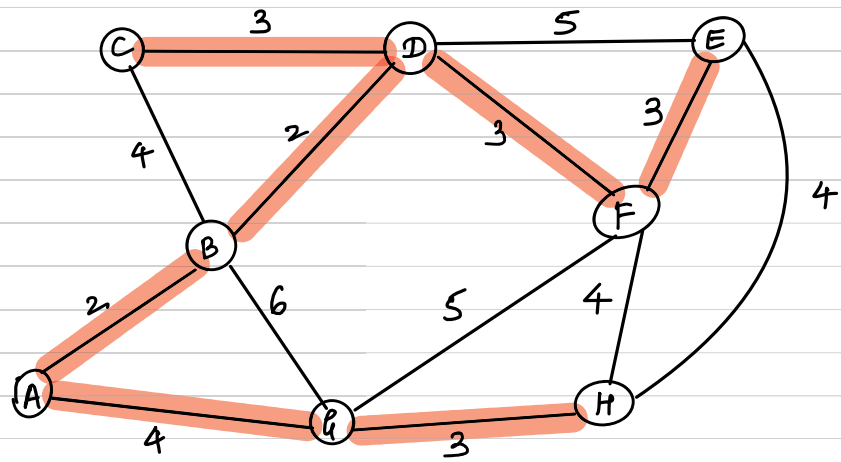
$W(T) = 2+2+3+4+3+3+3$

$= 20.$

(3) Find the minimal spanning tree of the following graph using Prims algorithm.



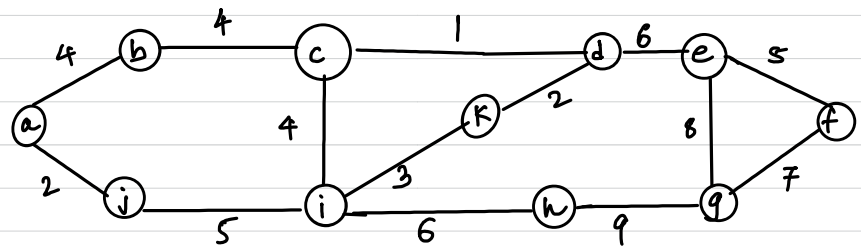
Sol^n :



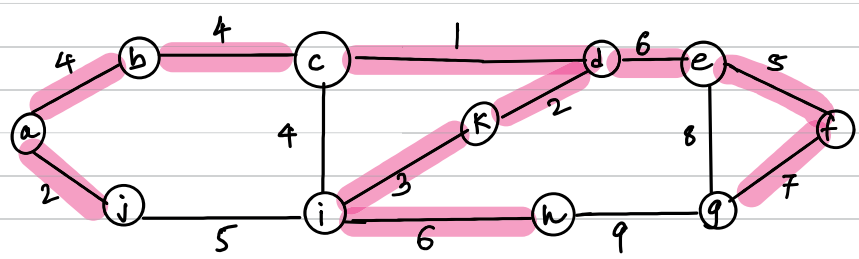
$W(T) = 3 + 2 + 2 + 4 + 3 + 3 + 3$

$= 20$

(4) Find the minimal spanning tree using Kruskal Algorithm.



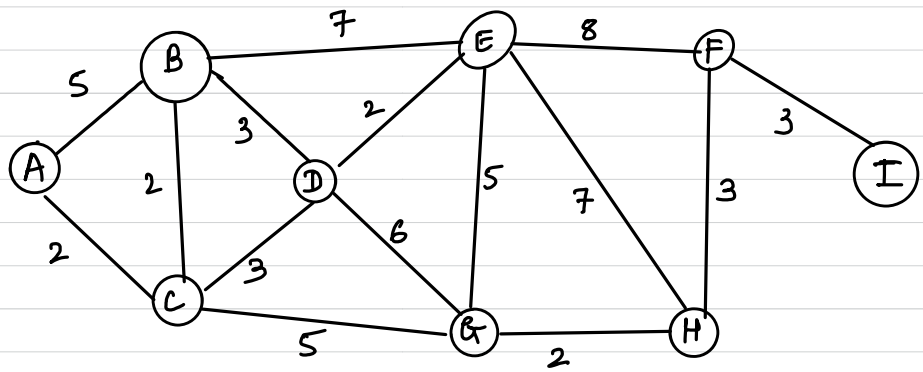
Sol^n :



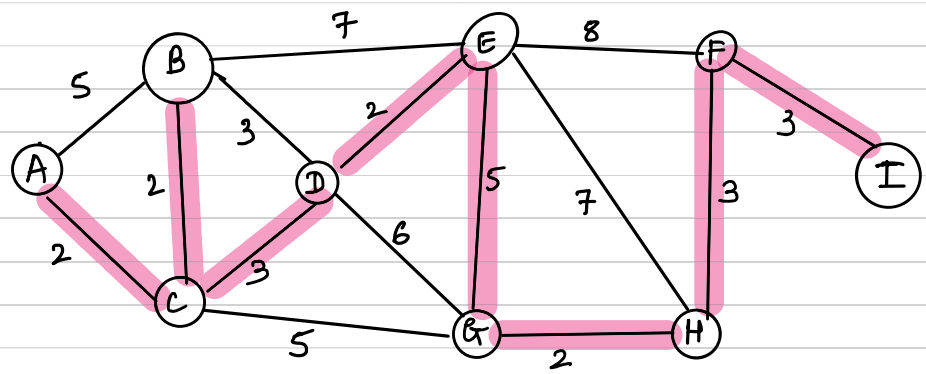
$W(T) = 2 + 4 + 4 + 1 + 3 + 2 + 6 + 6 + 5 + 7$

$= 40$

(5) Find minimal spanning tree using Prims and Kruskal Algorithm.



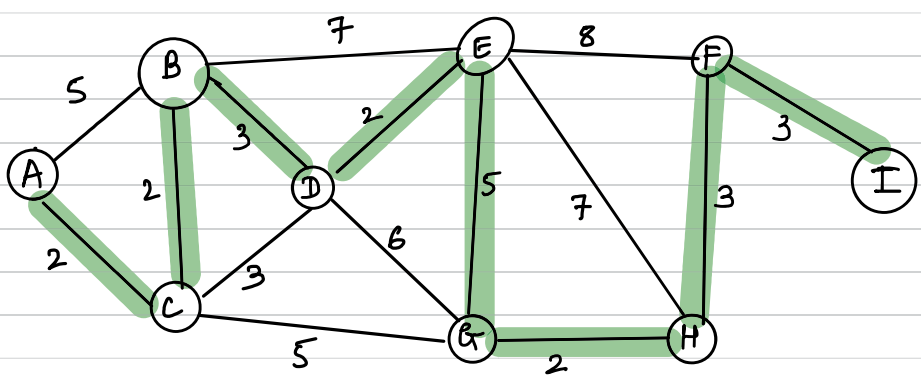
Sol^n : Prims Algorithm



$W(T) = 2+2+3+2+5+2+3+3$

$= 22$

Kruskal Algorithm



$W(T) = 2+2+3+2+5+2+3+3 = 22.$

Discrete Mathematics Quiz Schedule :

Unit -4	27/11/2020	9:30 A.m to 10:00 A.m
Unit-1 to 4 (combined)	11/12/2020	9:30 A.m